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Important concepts, symbols, and equations

- A configuration can be represented by **exponential coordinates** $S\theta \in \mathbb{R}^6$: a screw axis S multiplied by the distance θ it is followed. (Equivalently, $\forall t$: a twist $\mathcal{V}$ and a time t it is followed.)
- As with rotations, we can define a matrix exponential and its inverse, the matrix log. The exponential “integrates a twist” for time 1, and the log finds the constant twist needed to achieve the displacement in time 1.

$$\begin{aligned} \exp : [S]\theta \in se(3) &\rightarrow T \in SE(3) \\ \log : T \in SE(3) &\rightarrow [S]\theta \in se(3) \quad \theta \in [0, \pi] \end{aligned}$$

Important concepts, symbols, and equations

For $S = (\omega, v)$, either

- $\|\omega\| = 1$:

$$e^{[S]\theta} = \begin{bmatrix} e^{[\omega]\theta} & (I\theta + (1 - \cos \theta)[\omega] + (\theta - \sin \theta)[\omega]^2) v \\ 0 & 1 \end{bmatrix}$$

- or $\omega = 0$ and $\|v\| = 1$:

$$e^{[S]\theta} = \begin{bmatrix} I & v\theta \\ 0 & 1 \end{bmatrix}$$

Important concepts, symbols, and equations (cont.)

- A **wrench** is $\mathcal{F} = (m, f) \in \mathbb{R}^6$. A linear force $f \in \mathbb{R}^3$ at r creates a moment $m = r \times f$.
- The dot product of a wrench and a twist is power: $P = \mathcal{V}^T \mathcal{F}$.
- The same wrench can be expressed in $\{a\}$ and $\{b\}$ as $\mathcal{F}_a$ and $\mathcal{F}_b$.
- Changing the frame of representation (power better be independent of the frame we use to represent twists and wrenches!):

$$\mathcal{V}_b^T \mathcal{F}_b = \mathcal{V}_a^T \mathcal{F}_a$$

$$\begin{aligned}\mathcal{V}_b^T \mathcal{F}_b &= ([\text{Ad}_{T_{ab}}] \mathcal{V}_b)^T \mathcal{F}_a \\ &= \mathcal{V}_b^T [\text{Ad}_{T_{ab}}]^T \mathcal{F}_a.\end{aligned}$$

$$\mathcal{F}_b = [\text{Ad}_{T_{ab}}]^T \mathcal{F}_a$$

Rotations	Rigid-Body Motions
$R \in SO(3) : 3 \times 3$ matrices $R^T R = I, \det R = 1$	$T \in SE(3) : 4 \times 4$ matrices $T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix},$ where $R \in SO(3), p \in \mathbb{R}^3$
$R^{-1} = R^T$	$T^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix}$
change of coordinate frame: $R_{ab}R_{bc} = R_{ac}, \quad R_{ab}p_b = p_a$	change of coordinate frame: $T_{ab}T_{bc} = T_{ac}, \quad T_{ab}p_b = p_a$

rotating a frame $\{b\}$:

$$R = \text{Rot}(\hat{\omega}, \theta)$$

$$R_{sb'} = RR_{sb}:$$

rotate θ about $\hat{\omega}_s = \hat{\omega}$

$$R_{sb''} = R_{sb}R:$$

rotate θ about $\hat{\omega}_b = \hat{\omega}$

displacing a frame $\{b\}$:

$$T = \begin{bmatrix} \text{Rot}(\hat{\omega}, \theta) & p \\ 0 & 1 \end{bmatrix}$$

$T_{sb'} = TT_{sb}$: rotate θ about $\hat{\omega}_s = \hat{\omega}$
(moves $\{b\}$ origin), translate p in $\{s\}$

$T_{sb''} = T_{sb}T$: translate p in $\{b\}$,
rotate θ about $\hat{\omega}$ in new body frame

unit rotation axis is $\hat{\omega} \in \mathbb{R}^3$,

where $\|\hat{\omega}\| = 1$

“unit” screw axis is $\mathcal{S} = \begin{bmatrix} \omega \\ v \end{bmatrix} \in \mathbb{R}^6$,

where either (i) $\|\omega\| = 1$ or
(ii) $\omega = 0$ and $\|v\| = 1$

for a screw axis $\{q, \hat{s}, h\}$ with finite h ,

$$\mathcal{S} = \begin{bmatrix} \omega \\ v \end{bmatrix} = \begin{bmatrix} \hat{s} \\ -\hat{s} \times q + h\hat{s} \end{bmatrix}$$

angular velocity is $\omega = \hat{\omega}\dot{\theta}$

twist is $\mathcal{V} = \mathcal{S}\dot{\theta}$

for any 3-vector, e.g., $\omega \in \mathbb{R}^3$,

$$[\omega] = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \in so(3)$$

identities, $\omega, x \in \mathbb{R}^3, R \in SO(3)$:

$$\begin{aligned} [\omega] &= -[\omega]^T, [\omega]x = -[x]\omega, \\ [\omega][x] &= ([x][\omega])^T, R[\omega]R^T = [R\omega] \end{aligned}$$

$$\dot{R}R^{-1} = [\omega_s], \quad R^{-1}\dot{R} = [\omega_b]$$

$$\text{for } \mathcal{V} = \begin{bmatrix} \omega \\ v \end{bmatrix} \in \mathbb{R}^6,$$

$$[\mathcal{V}] = \begin{bmatrix} [\omega] & v \\ 0 & 0 \end{bmatrix} \in se(3)$$

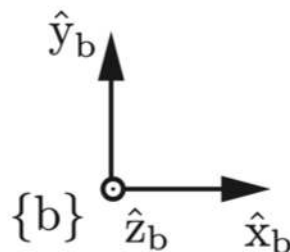
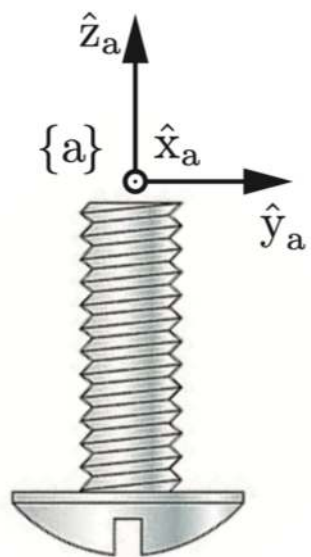
(the pair (ω, v) can be a twist $\mathcal{V}$
or a “unit” screw axis $\mathcal{S}$,
depending on the context)

$$\dot{T}T^{-1} = [\mathcal{V}_s], \quad T^{-1}\dot{T} = [\mathcal{V}_b]$$

$$[\text{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

identities: $[\text{Ad}_T]^{-1} = [\text{Ad}_{T^{-1}}],$
 $[\text{Ad}_{T_1}][\text{Ad}_{T_2}] = [\text{Ad}_{T_1 T_2}]$

change of coordinate frame: $\hat{\omega}_a = R_{ab}\hat{\omega}_b, \quad \omega_a = R_{ab}\omega_b$	change of coordinate frame: $\mathcal{S}_a = [\text{Ad}_{T_{ab}}]\mathcal{S}_b, \quad \mathcal{V}_a = [\text{Ad}_{T_{ab}}]\mathcal{V}_b$
exp coords for $R \in SO(3)$: $\hat{\omega}\theta \in \mathbb{R}^3$	exp coords for $T \in SE(3)$: $\mathcal{S}\theta \in \mathbb{R}^6$
exp : $[\hat{\omega}]\theta \in so(3) \rightarrow R \in SO(3)$ $R = \text{Rot}(\hat{\omega}, \theta) = e^{[\hat{\omega}]\theta} =$ $I + \sin \theta [\hat{\omega}] + (1 - \cos \theta) [\hat{\omega}]^2$	exp : $[\mathcal{S}]\theta \in se(3) \rightarrow T \in SE(3)$ $T = e^{[\mathcal{S}]\theta} = \begin{bmatrix} e^{[\omega]\theta} & * \\ 0 & 1 \end{bmatrix}$ where $*$ = $(I\theta + (1 - \cos \theta)[\omega] + (\theta - \sin \theta)[\omega]^2)v$
log : $R \in SO(3) \rightarrow [\hat{\omega}]\theta \in so(3)$ algorithm in Section 3.2.3.3	log : $T \in SE(3) \rightarrow [\mathcal{S}]\theta \in se(3)$ algorithm in Section 3.3.3.2
moment change of coord frame: $m_a = R_{ab}m_b$	wrench change of coord frame: $\mathcal{F}_a = (m_a, f_a) = [\text{Ad}_{T_{ba}}]^T \mathcal{F}_b$



A screw axis is defined by the screw image (positive motion drives the screw upward), and the pitch is 5 mm/rad. The origin of {b} is at (0,4,-2) mm in {a}.

What is T_{ab} ?

$$T_{ab} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

What is the screw S_a ? S_b ?

$$S_a = (0, 0, 1, 0, 0, 5)$$

$$S_b = (0, 1, 0, 0, 5, -4)$$

If {b} follows the screw a distance θ , what is the mathematical expression for the final configuration $T_{ab'}$?

$$T_{ab'} = T_{ab} \exp([S_b]\theta) \text{ or } T_{ab'} = \exp([S_a]\theta) T_{ab}$$

If $\theta = \pi$, give the numerical entries of $T_{ab'}$.

$$T_{ab'} = \begin{bmatrix} 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 5\pi - 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ by inspection}$$

Given frames $\{a\}$, $\{b\}$, and $\{c\}$, and their representations relative to each other T_{ab} and T_{ac} , write the twist needed to move $\{b\}$ to $\{c\}$ in t seconds in the $se(3)$ form $[V_a]$.

$$T_{ac} = \exp([V_a]t) T_{ab}$$

$$T_{ac} T_{ab}^{-1} = \exp([V_a]t)$$

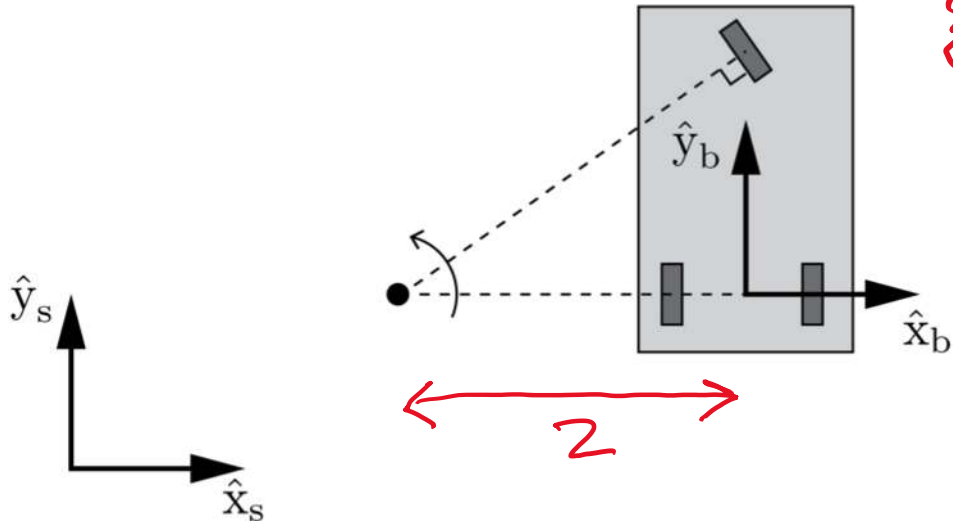
$$\log(T_{ac} T_{ba}) = [V_a]t$$

$$\frac{1}{t} \log(T_{ac} T_{ba}) = [V_a]$$

Car {b} frame origin is initially at (4,1,0) in {s} and it drives at a constant steering angle with a turning radius of 2. What is the screw axis $(q, \hat{s}, h)$ expressed in {b}? {s}?

$$\{b\}: q = (-2, 0, 0) \quad \hat{s} = (0, 0, 1) \quad h = 0$$

$$\{s\}: q = (2, 1, 0) \quad \hat{s} = (0, 0, 1) \quad h = 0$$



What is the screw S_b ? S_s ?

$$S_b = (0, 0, 1, 0, 2, 0)$$

$$S_s = (0, 0, 1, 1, -2, 0)$$

If the car's forward speed is 4, what is v_b ? v_s ?

$$\text{linear speed} = r \dot{\theta}$$

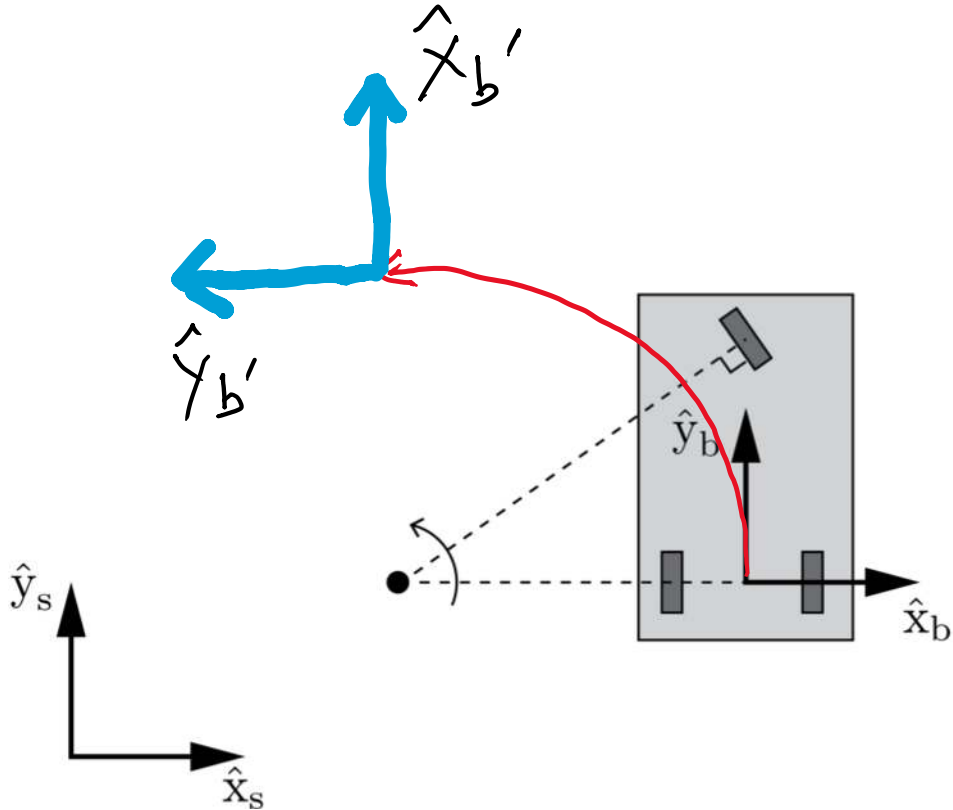
$$4 = 2 \dot{\theta} \quad \dot{\theta} = 2$$

$$v_b = 2 S_b$$

$$v_s = 2 S_s$$

If the car completes a quarter of a rotation, what are the exponential coordinates $S_b\theta$? $S_s\theta$?

$$\theta = \frac{\pi}{2}, \text{ so } \frac{\pi}{2} \mathcal{J}_b \text{ and } \frac{\pi}{2} \mathcal{J}_s$$



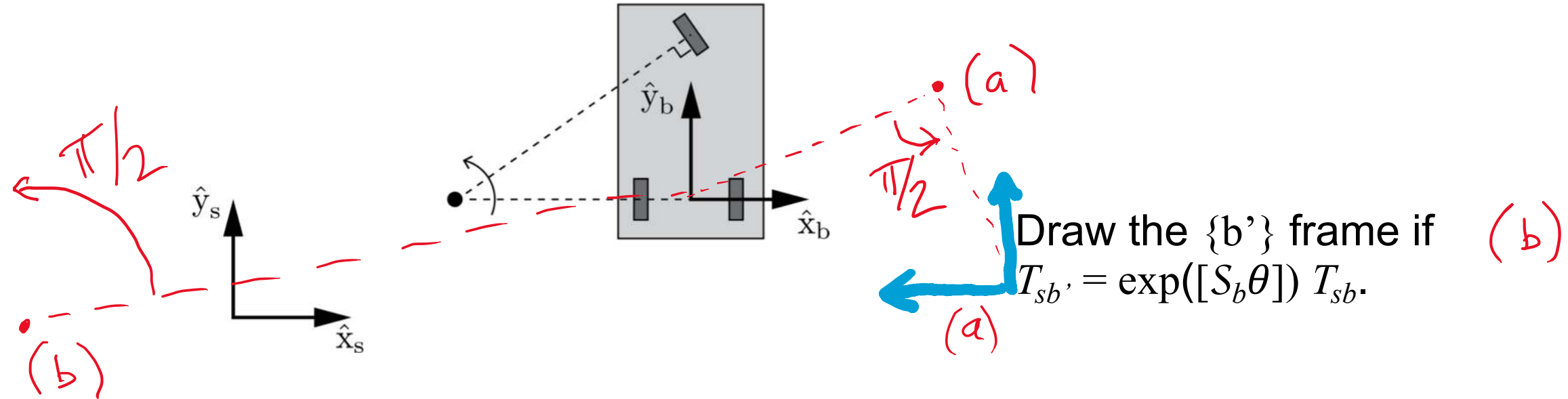
Where does the car end up? Draw a picture.

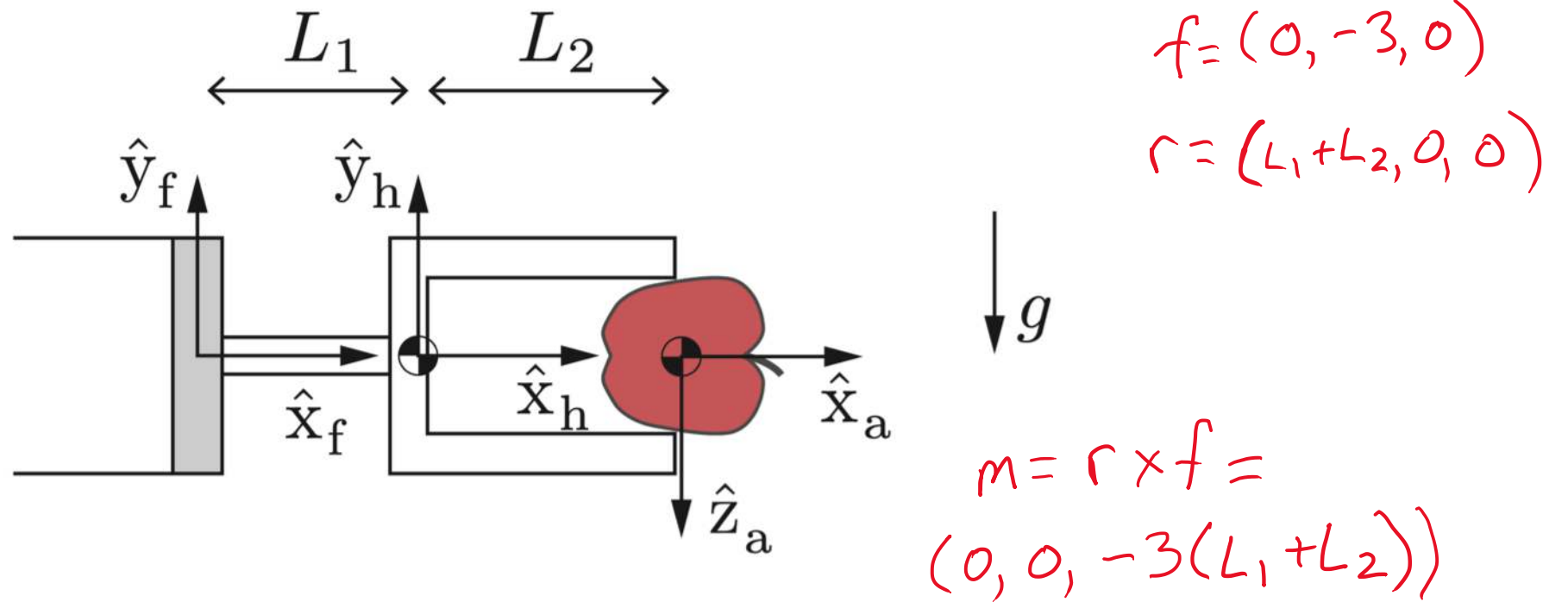
Express this final configuration mathematically, in terms of T_{sb} (as shown in the figure) and (1) the matrix exponential of $[S_b\theta]$ or (2) the matrix exponential of $[S_s\theta]$.

$$T_{sb}' = \exp([\mathcal{J}_s] \theta) T_{sb} \quad \text{or}$$

$$T_{sb}' = T_{sb} \exp([\mathcal{J}_b] \theta)$$

(a) Draw the $\{b'\}$ frame if
 $T_{sb'} = T_{sb} \exp([S_s \theta])$.





If gravity acting on the apple causes a downward force of 3 N, what is the wrench $\mathcal{F}_f$ felt at the force-torque sensor due to the apple?

$$\mathcal{F}_f = (0, 0, -3(L_1 + L_2), 0, -3, 0)$$